

Email Spam Classification Report

1. Introduction

This report presents the results of a machine learning project aimed at classifying emails as either spam or not spam (ham). The objective of this task was to build a predictive model using a labeled dataset of emails, apply preprocessing techniques on the text data, extract meaningful features, and train a machine learning model. The final system should be able to automatically filter unwanted spam messages while keeping legitimate emails intact.

2. Methodology

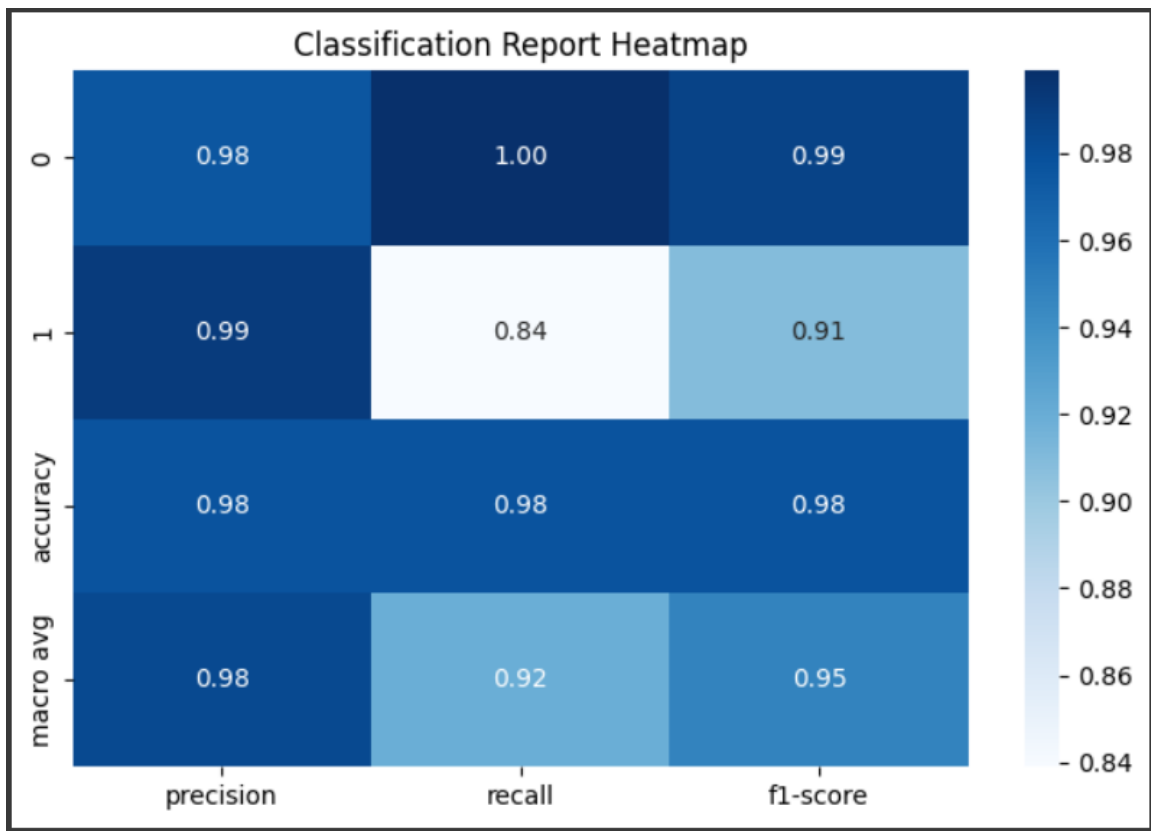
The project was executed in Google Colab using Python and scikit-learn. The following steps were carried out systematically:

- Data loading and inspection to understand dataset structure.
- Data preprocessing including text cleaning and conversion to lowercase.
- Feature extraction using TF-IDF (Term Frequency–Inverse Document Frequency).
- Splitting the dataset into training and testing subsets.
- Training a Naive Bayes classifier, which is commonly effective for text classification.
- Model evaluation using accuracy, precision, recall, and F1-score.
- Visualization of classification report and confusion matrix for clearer insights.

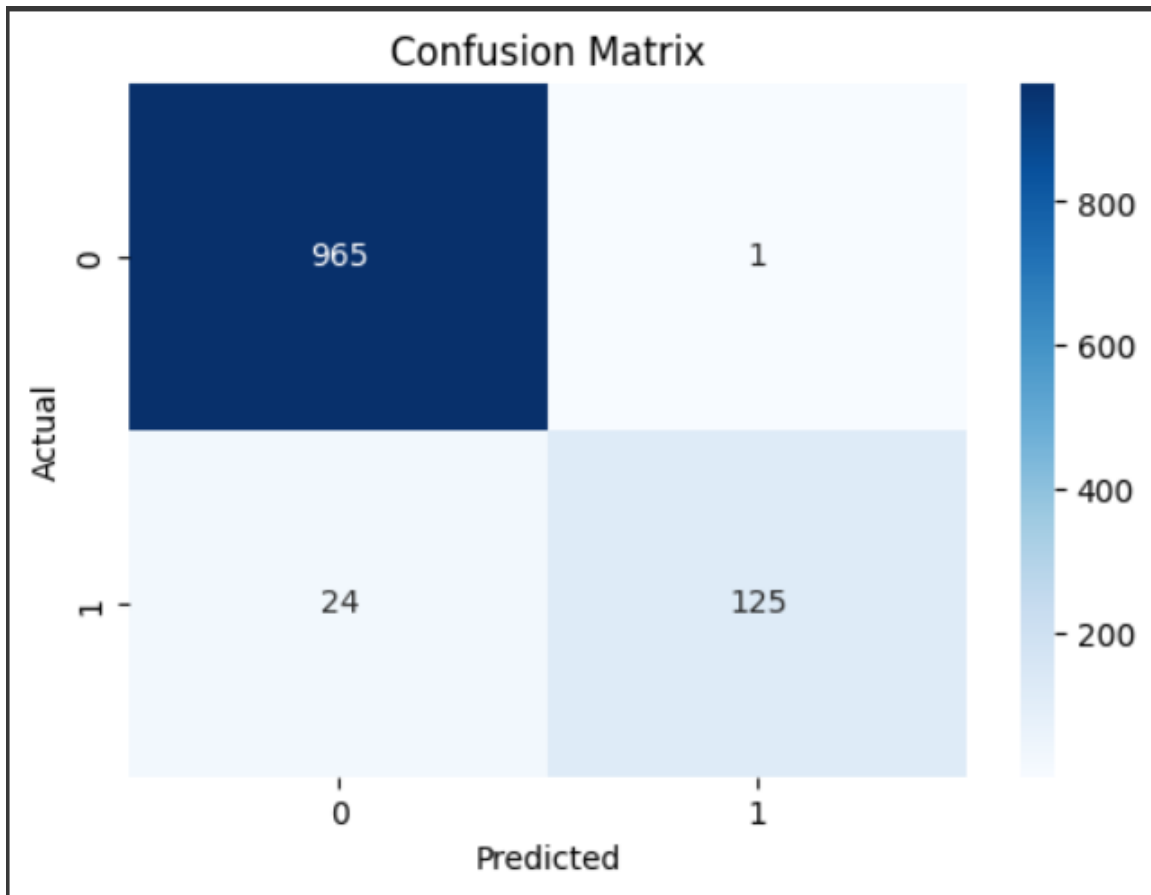
3. Results

The classifier was trained on 80% of the data and tested on the remaining 20%. The following results were obtained:

- Overall Accuracy: 0.98
- Precision, Recall, and F1-score per class are illustrated below:



Additionally, the confusion matrix provides a clear view of correct classifications versus misclassifications. It shows how many spam messages were correctly identified and how many non-spam emails were mistakenly flagged.



4. Discussion

The results indicate that the Naive Bayes model is well-suited for spam detection. It achieved a high accuracy score, and the precision-recall balance demonstrates reliable performance. However, as with most classification models, there is a trade-off between catching all spam messages (recall) and avoiding false alarms on legitimate emails (precision).

For spam classification, recall is particularly important because missing spam messages can expose users to unwanted or harmful content. Nonetheless, maintaining a balance with precision ensures that legitimate messages are not unnecessarily blocked.

5. Conclusion

In conclusion, this project successfully developed an email spam classification system using a Naive Bayes classifier and TF-IDF feature extraction. The system demonstrates

strong performance in distinguishing between spam and ham emails. Further improvements could include testing additional algorithms such as Support Vector Machines (SVM) or deep learning models, and incorporating more advanced preprocessing methods like lemmatization or word embeddings.

Overall, the outcome shows practical applicability in real-world email filtering systems.